RESERVEPARK

SoC

Smart Reservation + Auto Bay Recovery

License plate access • 15-minute grace • 45-minute no-show policy
Auto bay allocation • Protected backup pool • Sensor-verified recovery



HIGH-LEVEL SoC ARCHITECTURE SPECIFICATION (HAS)

Design objectives & project scope

A booking-first SoC that verifies both authorization and physical parking status

Mandatory SoC requirements

- At least one counter; minimum two input units and two output units.
- Built-in register file and memory for reservations, records and logs.
- An exit FSM consistent with the architecture.
- Directed and random validation; hardware-oriented DFx.
- At least one advanced / smart feature.

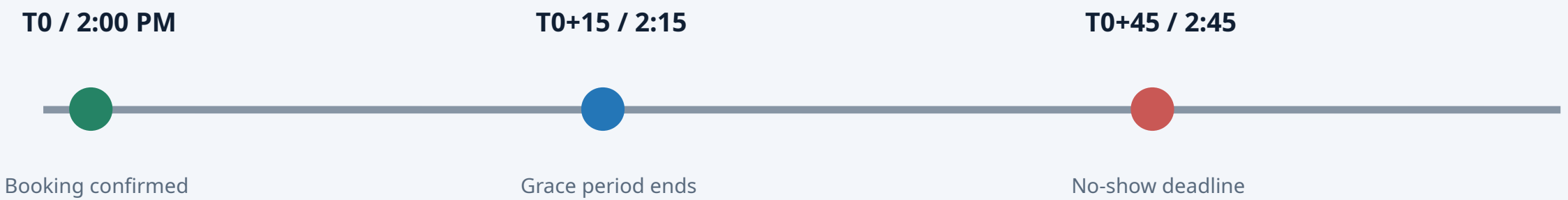
Our key design choices

- 100% reservation-only; no walk-in access.
- Automatically assign a numbered bay for each booking.
- Protect a configurable pool of backup bays (e.g., 5/100).
- 15-minute grace; no-show at 45 minutes if not checked in.
- If assigned bay is blocked, reassign before admission.
- Cameras, sensors and counters cross-check physical status.

Scope boundary: Mobile app, cloud reservation service and payment gateway are EXTERNAL; SoC controls on-site real-time decisions.

Booking policy: 15-minute grace + 45-minute no-show

Example: booking confirmed at 2:00 PM (T0); arrival countdown begins immediately



0–15 min | ON TIME

Valid plate + arrival check-in. No penalty.

15–45 min | LATE

Booking remains held; app warning sent.

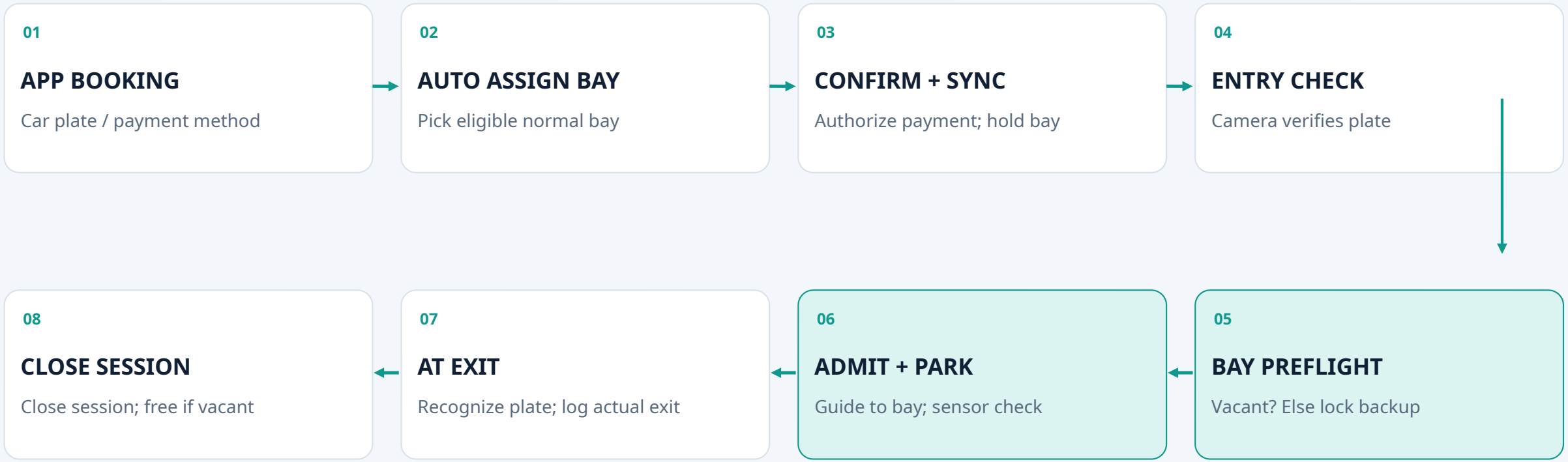
AT +45 min | NO SHOW

Release reserved bay; request penalty once.

Booking starts the timer. Gate-verified check-in stops the no-show clock; parking has NO departure deadline.

End-to-end operation

Reservation → authentication → sensor-confirmed parking → exit



PARALLEL TIMER: If no verified check-in by BOOKING T0+45 → NO_SHOW → release bay hold → request one fee.

Auto-assign a parking bay at the time of booking

Protected backup spaces are never sold as ordinary reservations

1 REQUEST

Plate number + payment authorisation

App sends booking request

2 FILTER

NORMAL bay must be vacant + unheld

Check live bay sensors and health

3 CHOOSE

Pick a compatible eligible bay

Choose by compatibility / lowest ID

4 COMMIT

Lock bay + unique booking ID

Payment authorization; atomic reservation

5 CONFIRM

Bay A12 shown in app

Sync assigned bay to local SoC

100-BAY EXAMPLE

95

NORMAL BAYS

5

PROTECTED BACKUPS

Backup bay only used by an already confirmed reservation when its assigned bay fails preflight.

Concurrency rule: atomically lock one vacant NORMAL bay per booking; never assume a parked vehicle will leave at a predicted time.

Occupancy counting: physical ≠ reserved ≠ bookable

Bookable bays are vacant, healthy, unheld NORMAL bays — no predicted departure required

100	30	20	5	2	43
TOTAL BAYS	OCCUPIED	EMPTY + HELD	BACKUP EMPTY	FAULT / BLOCKED	BOOKABLE NOW

Four key inventory quantities

- Occupied: sensors confirm parked vehicles.
- Held: empty bays reserved for active bookings.
- Backup-free: protected, vacant and healthy bays.
- Bookable: vacant NORMAL bays right now.

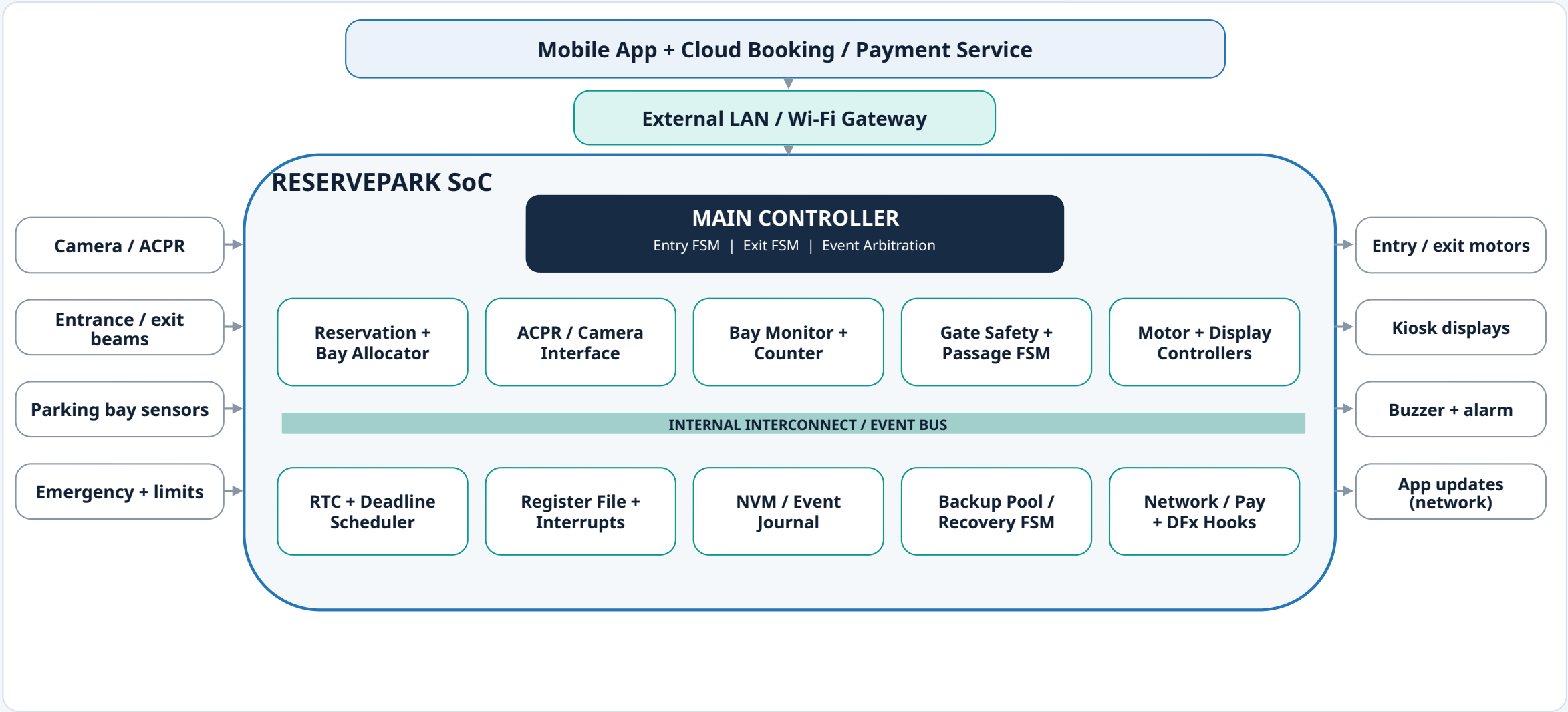
Key invariants

- $0 \le \text{occupied} \le \text{total}$; backup-free ≤ quota.
- Never promise an occupied or held bay.
- One active reservation → one current bay ID.
- Do not derive capacity by subtracting overlapping counters.

Example categories are mutually exclusive. A parked bay remains unavailable until physically vacant; future checkout time is unknown.

4.1 | Proposed top-level SoC block diagram

External app/payment are outside chip; each internal block is on the SoC interconnect



4.2 | Subsystems and IP functionality

ACPR, barrier motor controller and display controller may be treated as black boxes

Subsystem / IP	Function
Main Controller	Runs entry/exit FSMs; arbitrates events, faults and interrupts.
Booking & Bay Allocator	Matches plate; assigns a physically vacant NORMAL bay at booking.
Backup Pool & Recovery FSM	Maintains protected backup bays; locks and swaps assignments before entry.
Camera / ACPR*	Converts camera frames into plate ID, confidence and valid/error status.
RTC + Deadline Timer	Measures 15/45 minutes from booking T0; expires unarrived holds.
Bay Monitor & Counter	Tracks occupancy, reserved holds, free backups and bookable normal bays.
Gate Safety / Passage	Checks beams/limits; confirms crossing; inhibits unsafe closing.
Barrier Motor Controller*	Drives entry/exit actuator, receives opened/closed acknowledgements.
Display Controller*	Shows reserved/denied/full, assigned bay and fault instructions.
Memory / Journal Manager	Stores reservations, session events, timestamps and recovery records.
Network / Payment Events	Syncs bookings, app notices and actual-duration fee events.
Register / IRQ / DFx	Exposes status/configuration, faults, trace and test controls.

4.3 | Complete external signal interface list

Direction is defined from the SoC perspective

INPUT SIGNAL	DIR	DESCRIPTION
camera_frame / acpr_result	IN	Plate result + validity from camera path
entry_presence / exit_presence	IN	Car approaching a gate
entry_beam_A/B, exit_beam_A/B	IN	Detect ordered physical crossings
bay_occupied[0:N−1]	IN	Occupancy of each numbered parking bay
barrier_opened / barrier_closed	IN	Actuator limit-switch feedback
gate_obstruction	IN	Detect vehicle/person in barrier path
emergency_button	IN	Request help / operator override
cloud_booking_rx / payment_ack	IN	Plate, booking ID, booking T0 and payment acknowledgment
bay_policy_cfg_rx	IN	Normal/backup bay map and capacity policy

OUTPUT SIGNAL	DIR	DESCRIPTION
entry_open / entry_close	OUT	Entry barrier motor control request
exit_open / exit_close	OUT	Exit barrier motor control request
entry_display / exit_display	OUT	Instruction + allocated bay + error
alarm / status_led	OUT	Warning, fault, emergency indication
cloud_status_tx	OUT	Verified entry/exit timestamps, bay state and occupancy
payment_request_tx	OUT	No-show penalty or actual-duration fee request
booking_ack_tx	OUT	Auto-assigned bay ID + confirmation / refusal
app_notification_tx	OUT	Late / no-show / reassigned-bay update
bay_recovery_tx	OUT	Backup lock/swap confirmation or no-space fault

4.4.1 | Non-volatile memory and transactional records

Persist bookings + committed movements even when power is interrupted

PERSISTED DATA	WHY IT IS NEEDED
Booking ID + original/current bay	Auto-assignment and any recovery transfer of the same booking
License plate token / encrypted ID	Match ACPR result to booked vehicle; avoid raw plaintext when possible
Booking / check-in / entry / exit timestamps	Time of booking T0, verified arrival and actual departure; 15/45-minute deadlines
Booking state	HELD / ARRIVAL_PENDING / PARKED / NO_SHOW / CLOSED
Actual entry / exit timestamps	No predetermined end time; compute duration from confirmed passage events
Payment transaction ID + status	Track fee request / acknowledgement; protect against repeat charge
Occupancy snapshot + event sequence	Restart reconciliation with latest committed events
Bay role / health / current hold	Normal/backup role, compatibility, current exclusive hold and sensor health
Reassignment event + lock ID	Preserve A12 → B03 swap, reason, unique commit and recovery audit
Fault / audit logs	Sensor mismatch, bay unavailable, gate timeout, operator action

Journal assignments and movements atomically. On restart, reconcile live sensors; no guessed departure or repeated fees.

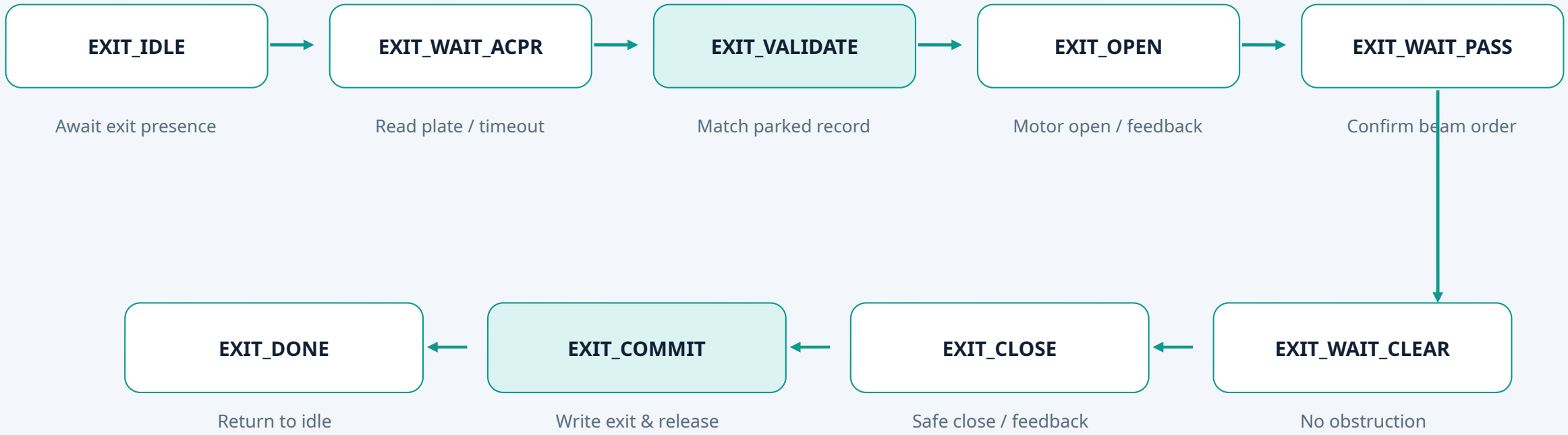
4.4.2 | Register-file map

Memory-mapped control/status registers (field-by-field definitions in appendix)

REGISTER	BITS	FIELD PURPOSE
R0 SYSTEM_CTRL	16	Enable, maintenance, booking enable, alerts
R1 SYSTEM_STATUS	16	Network, NVM, RTC ready; busy/fault state
R2 COUNT_STATUS	16	[7:0] physically occupied; [15:8] bookable now
R3 RESERVATION_COUNT	16	[7:0] reserved not arrived; [15:8] in transit
R4 TIME_POLICY	16	Grace minutes = 15; no-show minutes = 45
R5 CAPACITY_CFG	16	Total bay count; max 255 in demo
R6 GATE_COMMAND	16	Entry/exit open/close command pulses
R7 GATE_STATUS	16	Gate limit feedback, beams, passage events
R8 BOOKING_STATUS	16	Selected booking match, grace/late flags and no-show status
R9 IRQ_STATUS / R10 IRQ_ENABLE	16 each	Sticky fault causes and interrupt masks
R11 DFX_CTRL	16	Self-test, trace, secured input injection
R12 RTC_SECONDS	32	Trusted clock; used for arrival deadlines and actual stay duration
R13 BACKUP_POOL_CFG / R14 STATUS	16 each	Protected quota; backup-free and backup-in-use counts
R15 ALLOC_COMMAND / R16 STATUS	16 each	Assign/reassign/lock; success/conflict/failure flags
R17 BAY_ID_SNAPSHOT	16	Current bay [7:0] and original bay [15:8]

4.5 | Exit finite-state machine (normal path)

No departure deadline; record actual exit and bill for actual parked duration



Confirmed passage sets the actual exit timestamp. Release a bay only after sensor-confirmed vacancy. Sensor conflicts trigger review, not an assumed departure.

4.5 | FSM exception paths and state descriptions

All failures preserve safe motor behavior, occupancy integrity and a recoverable audit trail

STATE	ACTION / TRANSITION
EXIT_IDLE	Wait for exit presence; exit barrier normally closed.
EXIT_WAIT_ACPR	Trigger recognition; retry or timeout on invalid result.
EXIT_VALIDATE	Find active parked record; reject/route unmatched plate to assistance.
EXIT_OPEN	Issue opening command; await opened acknowledgement or timeout.
EXIT_WAIT_PASS	Check correct directional sensor sequence; deduplicate repeats.
EXIT_WAIT_CLEAR	Keep open while lane occupied; report timeout if obstruction persists.
EXIT_CLOSE	Close only if safety clear; await closed acknowledgement.
EXIT_COMMIT	Persist actual exit time once; update occupancy and bookability after vacancy.
EXIT_DONE	Notify cloud/app; return to idle.
EXIT_FAULT / ASSIST	ACPR, unmatched booking, beam, motor or memory failure; flag operator.

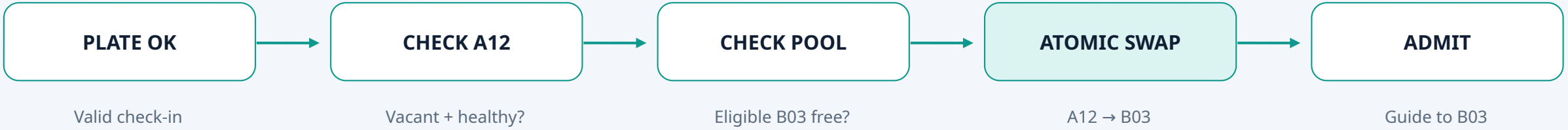
Entry access and no-show scheduler

Additional operational logic keeps the exit FSM and reservation policy consistent

LOGICAL STATE / EVENT	ACTION
AUTO_ASSIGNED	App booking accepted → select one physically vacant NORMAL bay; hold it.
GRACE / LATE	From booking T0: 0–15 minute grace; retain hold until +45 minutes.
AT ENTRY	Plate and entrance sensor verify an active booking; log check-in.
BAY_PREFLIGHT	Check assigned bay sensor/health; if blocked, request protected backup.
REASSIGN_LOCK	Atomically lock compatible backup, swap bay ID; notify app/kiosk.
ARRIVAL_PENDING	Check-in accepted; blocked gate does not trigger no-show charge.
ENTRY_CONFIRMED	Beams confirm physical entry; bay sensor confirms parked status.
NO_SHOW	No check-in at BOOKING T0+45: cancel hold; send fee event once.
MANUAL_REVIEW	No backup or gate error: provide assistance/refund; no unfair fee.

Backup-bay recovery | keep the reservation valid

Reassign BEFORE opening the entrance barrier; no walk-in spaces are created



ASSIGNED BAY IS FREE

Skip recovery. Keep original A12 if physically vacant, healthy and exclusively held; open the barrier when safe.

NO SUITABLE BACKUP

DO NOT ADMIT. Keep arrival verified, inform app/kiosk, raise REASSIGN_FAIL, offer assistance or cancellation/refund. Never apply no-show fee.

Recovery guards Only VACANT, HEALTHY and unheld compatible backup bays qualify. Commit the swap atomically; A12 stays blocked until vacant and verified.

4.6 | Directed + randomized validation plan

Each test specifies a stimulus and an observable expected result

DIRECTED TESTS

ID	STIMULUS	EXPECTED
D1	Book with plate; no departure time	Exactly one vacant NORMAL bay assigned; booking T0 stored.
D2	Two simultaneous bay booking requests	Atomic locks; never issue same live bay to two bookings.
D3	T0+12 and T0+30 plate arrivals	Both accepted; latter marked late, not charged.
D4	No verified arrival at T0+45	Release hold; one idempotent no-show charge.
D5	Assigned A12 occupied; B03 backup free	Lock B03; update booking/app; admit to B03.
D6	Assigned A12 blocked; no backup free	Keep closed; assistance/refund, never no-show fee.
D7	Leave backup B03 after 12 hours	No overstay error; log actual exit; free B03 only when vacant.
D8	Power reset during A12→B03 swap	Journal replay; single current bay, no double hold.

RANDOM TESTS

ID	RANDOM CONDITION	PASS CRITERION
R1	Concurrent bookings + live bay contention	One owner per bay; no assumed future departures.
R2	Random bay sensor faults / blocked backups	Never assign an occupied or unhealthy backup.
R3	Random resets, booking ages and network loss	Exactly-once events; no unjust fee; ledger valid.

4.7 | Design for Excellence / Debug & Test

Use scan, hardware trace and self-test to locate hardware faults quickly

Issue A — sensor input stuck HIGH/LOW

- Symptom: normal or backup bay falsely appears free.
- Debug hook: raw/debounced sensor visibility, heartbeat and sticky SENSOR_ERR.
- DFx: secured input test pattern injection and GPIO path test.

Issue B — occupancy count stuck/corrupt

- Symptom: duplicate assignment or backup count mismatch.
- Debug hook: allocation ID, bay IDs, lock status + trace.
- DFx: scan observation, allocator self-test, memory BIST.

Ways to reduce hardware test time

- Run memory BIST and independent peripheral self-tests in parallel where isolation permits.
- Use scan chains, compressed test patterns and a trace buffer to make failures observable.
- Separate entry, exit, RTC, sensor and counter IP tests; only retest failed domains.

Safety rule: test injection is restricted to authorized test/maintenance mode; it cannot disable live gate obstruction interlocks.

Innovation: more than an ordinary parking counter

Each feature directly produces an IP block, a register field, an FSM behavior and a test

01

Automatic numbered-bay assignment

Assign a vacant NORMAL bay now with one exclusive lock and app confirmation.

02

Timed reservation management

15-minute grace; cancel after 45-minute no-show with a once-only fee event.

03

Protected backup-bay recovery

Preflight assigned bay; atomically reassign a free backup before admitting the car.

04

Sensor-verified, fault-safe counting

Plate + gate + bay sensors; actual stay billing and fault-safe event logs.

Judge story: “We give every booking a specific bay - and automatically recover when that promised bay is not usable.”

RESERVEPARK SoC

Every valid booking gets a bay - and a recovery plan.

WHO?

License plate + booking

WHEN?

15/45 arrival; flexible stay

WHERE?

Auto-assigned + backup bays

Thank you | Questions & Answers

Register details 1

Bit allocations, access types and reset values for all SoC control registers

R0 SYSTEM_CTRL — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
ENABLE	0	RW	1	Enable controller operation
MAINT_MODE	1	RW	0	Maintenance / supervised override
BOOKING_EN	2	RW	1	Allow app reservation matching
ALERT_EN	3	RW	1	Enable external alarm outputs
RSVD	15:4	RO	0	Reserved bits read as zero

Register details 2

Bit allocations, access types and reset values for all SoC control registers

R1 SYSTEM_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
NET_OK	0	RO	0	Link established
NVM_READY	1	RO	0	Recovered journal and storage ready
TIME_VALID	2	RO	0	Trusted RTC synchronization achieved
FAULT	3	RO	0	Any sticky active system fault
ENTRY_BUSY	4	RO	0	Entry FSM active
EXIT_BUSY	5	RO	0	Exit FSM active
COUNT_DIFF	6	RO	0	Sensor / event mismatch
OBSTRUCTED	7	RO	0	Either gate obstruction
RSVD	15:8	RO	0	Reserved

Register details 3

Bit allocations, access types and reset values for all SoC control registers

R2 COUNT_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
PHYS_OCC	7:0	RO	0	Debounced physically occupied bays
BOOKABLE_NOW	15:8	RO	0	Vacant, healthy and unheld NORMAL bay now

R3 RESERVATION_COUNT — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
HELD_NOT_ARRIVED	7:0	RO	0	Reserved, no confirmed arrival
IN_TRANSIT	15:8	RO	0	Checked-in vehicle not yet parked / final exited

R4 TIME_POLICY — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
GRACE_MIN	7:0	RW	15	Minutes after confirmed booking time T0
NO_SHOW_MIN	15:8	RW	45	No-check-in cutoff from booking time T0

Register details 4

Bit allocations, access types and reset values for all SoC control registers

R5 CAPACITY_CFG — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
TOTAL_BAYS	7:0	RW	0	Configured bay count; demo up to 255
RSVD	15:8	RO	0	Reserved bits read zero

R6 GATE_COMMAND — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
ENTRY_OPEN	0	WO/PULSE	0	Request entry open
ENTRY_CLOSE	1	WO/PULSE	0	Request entry close (safety checked)
EXIT_OPEN	2	WO/PULSE	0	Request exit open
EXIT_CLOSE	3	WO/PULSE	0	Request exit close (safety checked)
RSVD	15:4	WO	0	Write zero; ignored

Register details 5

Bit allocations, access types and reset values for all SoC control registers

R7 GATE_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
ENTRY_OPENED	0	RO	0	Entry open limit
ENTRY_CLOSED	1	RO	0	Entry closed limit
EXIT_OPENED	2	RO	0	Exit open limit
EXIT_CLOSED	3	RO	0	Exit closed limit
ENTRY_OBSTRUCT	4	RO	0	Entry obstruction
EXIT_OBSTRUCT	5	RO	0	Exit obstruction
ENTRY_PASSED	6	RO	0	Entry directional event pulse
EXIT_PASSED	7	RO	0	Exit directional event pulse
RSVD	15:8	RO	0	Reserved

Register details 6

Bit allocations, access types and reset values for all SoC control registers

R8 BOOKING_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
PLATE_MATCH	0	RO	0	Selected event matches a booking
ACTIVE_BOOKING	1	RO	0	Active booking, still within 45 min or already checked in
GRACE_ACTIVE	2	RO	0	Within 15-minute grace
LATE_ACTIVE	3	RO	0	After grace, before 45-minute cutoff
CHECKED_IN	4	RO	0	Arrival verified before deadline
NO_SHOW	5	RO	0	No-show state for selected booking
BAY_CONFIRMED	6	RO	0	Assigned bay sensor occupied
PAYMENT_ACK	7	RO	0	Payment provider acknowledged request
RECOVERY_USED	8	RO	0	Allocated backup for selected reservation
ASSIGN_VALID	9	RO	0	Selected booking has a committed current bay
RSVD	15:10	RO	0	Reserved

Register details 7

Bit allocations, access types and reset values for all SoC control registers

R9 IRQ_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
ACPR_ERR	0	RW/W1C	0	ACPR failure
COUNT_MISMATCH	1	RW/W1C	0	Sensor/event count difference
MOTOR_TIMEOUT	2	RW/W1C	0	Barrier movement timeout
LANE_BLOCK	3	RW/W1C	0	Barrier obstruction timeout
NO_SHOW_EVENT	4	RW/W1C	0	Deadline reached; no check-in
PAYMENT_ERR	5	RW/W1C	0	Fee request or ACK failure
NETWORK_ERR	6	RW/W1C	0	Lost cloud link
SENSOR_ERR	7	RW/W1C	0	Faulty sensor input
EMERGENCY	8	RW/W1C	0	Emergency request
BACKUP_LOW	9	RW/W1C	0	Protected free pool below policy threshold
REASSIGN_FAIL	10	RW/W1C	0	Backup missing or incompatible
BAY_CONFLICT	11	RW/W1C	0	Duplicate live hold or unexpected physical occupancy
RSVD	15:12	RO	0	Reserved

Register details 8

Bit allocations, access types and reset values for all SoC control registers

R10 IRQ_ENABLE — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
IRQ_MASK	11:0	RW	0	Enable individual IRQ_STATUS events
RSVD	15:12	RO	0	Reserved

R11 DFX_CTRL — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
START_MBIST	0	WO/PULSE	0	Trigger authorized memory BIST
TRACE_ENABLE	1	RW	0	Capture events in debug trace
SENSOR_TEST_EN	2	RW	0	Secured test-mode stimulus injection
RSVD	15:3	RO	0	Reserved

R12 RTC_SECONDS — 32-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
UNIX_TIME_S	31:0	RO	0	Time seconds; check R1.TIME_VALID before use

Register details 9

Bit allocations, access types and reset values for all SoC control registers

R13 BACKUP_POOL_CFG — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
BACKUP_QUOTA	7:0	RW	0	Provisioned protected bay count; example 5 of 100
RECOVERY_EN	8	RW	0	Enable only after bay map and NVM recovery are valid
RSVD	15:9	RO	0	Reserved

R14 BACKUP_POOL_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
BACKUP_FREE	7:0	RO	0	Healthy, unheld, vacant protected bays
BACKUP_IN_USE	15:8	RO	0	Backup bays locked or parked under recovery

Register details 10

Bit allocations, access types and reset values for all SoC control registers

R15 ALLOC_COMMAND — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
AUTO_ASSIGN	0	WO/PULSE	0	Choose available normal bay for new booking
BACKUP_REASSIGN	1	WO/PULSE	0	Atomic transfer to compatible protected bay
RELEASE_HOLD	2	WO/PULSE	0	Release an obsolete reservation hold
RSVD	15:3	WO	0	Write zero; ignored

R16 ALLOC_STATUS — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
ALLOC_OK	0	RO	0	New bay assignment committed
RECOVERY_USED	1	RO	0	Backup became current assignment
BACKUP_EXHAUSTED	2	RO	0	No suitable free protected bay
BAY_CONFLICT	3	RO	0	Slot held/occupied or incompatible
ASSIGN_LOCKED	4	RO	0	Atomic update in progress
RSVD	15:5	RO	0	Reserved

Register details 11

Bit allocations, access types and reset values for all SoC control registers

R17 BAY_ID_SNAPSHOT — 16-bit

FIELD	BIT	TYPE	RESET	DESCRIPTION
CURRENT_BAY_ID	7:0	RO	0	Current allocated bay; valid if R8.ASSIGN_VALID
ORIGINAL_BAY_ID	15:8	RO	0	Originally auto-assigned bay before recovery

Specification rationale & edge-case handling

Decisions that prevent inconsistent occupancy or unfair charges

EDGE / CORNER CASE	ROBUST ACTION
Driver provides no departure time	Assign a vacant NORMAL bay now; hold until check-in or no-show; never forecast checkout.
Backup pool is configured	Protected bays never join ordinary app inventory; free backup count is tracked.
Assigned bay occupied at arrival	Lock and reassign only a sensor-vacant backup; no use of estimated departures.
Backup exhausted or sensor unhealthy	Never admit with false promise; assistance/refund and no wrong no-show fee.
At T0+44 gate checks valid plate	Set ARRIVAL_PENDING before deadline; do not auto-charge because of queue/gate failure.
At T0+45 no arrival	Single NO_SHOW journal event; release bay; payment request with unique ID.
Sensor says occupied but no active booking	Flag unauthorized parking / discrepancy; bay not bookable.
Cloud disconnects	Use cached, verified bookings; suspend new remote booking requests; queue fee events.
Payment acknowledgment missing	Retry same transaction ID; never issue multiple distinct charges.
Car reverses after barrier open	Ignore opening alone; crossing requires A/B beams in correct order.
Bay sensor stays occupied after exit	Keep bay unavailable until confirmed vacant; flag sensor mismatch.
Power-on RTC invalid	Do not execute deadlines until trusted clock is restored.

Exit FSM transition guard conditions

Specific conditions used by RTL or a synthesizable control FSM

CURRENT STATE	GUARD / EVENT	NEXT STATE
EXIT_IDLE	exit_presence=1	EXIT_WAIT_ACPR
EXIT_WAIT_ACPR	acpr_valid=1	EXIT_VALIDATE
EXIT_WAIT_ACPR	acpr_timeout OR acpr_error	EXIT_FAULT / ASSIST
EXIT_VALIDATE	matched PARKED session	EXIT_OPEN
EXIT_VALIDATE	no valid active session	EXIT_FAULT / ASSIST
EXIT_OPEN	barrier_opened=1	EXIT_WAIT_PASS
EXIT_OPEN	motor_open_timeout	EXIT_FAULT / ASSIST
EXIT_WAIT_PASS	exit_A then exit_B sequence	EXIT_WAIT_CLEAR
EXIT_WAIT_PASS	wrong order / passage_timeout	EXIT_FAULT / ASSIST
EXIT_WAIT_CLEAR	lane_clear=1; do NOT wait for bay clearance	EXIT_CLOSE
EXIT_WAIT_CLEAR	obstruction=1	EXIT_WAIT_CLEAR (hold open)
EXIT_CLOSE	barrier_closed=1	EXIT_COMMIT
EXIT_CLOSE	closing_timeout / obstruction	EXIT_FAULT / ASSIST
EXIT_COMMIT	NVM ACK + once-only event with actual exit time	EXIT_DONE
EXIT_DONE	Cloud informed; actual duration sent for billing	EXIT_IDLE